Date ____ Day ____

GENERATIVE ADVERSIAL NETWORK

- Generative Models
  - ↳ Generative models are AI model that learns the underlying probability distribution or patterns of data and then generate new data samples similar to training data
  - ↳ Can generate
    - ↳ images
    - ↳ music
    - ↳ videos
    - ↳ speech
  - ↳ Learn, $P(X,Y)$ or directly $P(X)$

- Discriminative Model
  - ↳ Focuses on distinguishing b/w diff kinds of data
  - ↳ ~~Don't~~ They learn the boundaries for the speration of data
  - ↳ They ~~learn~~ don't learn how data is generated

Date ____ Day ____

- **Benefits Of Generative Model**

① Data Augmentation

↳ MRI Scans expensive, GAN generates additional scans

② Anomaly Detection

↳ Model learns "normal" data distribution, any thing "unusual" becomes anomaly

↳ Fraud Detection

③ Personalization

↳ Generated content on user preferences

④ Cost Efficiency

↳ Automates → Content Creation, research processes etc

- **Limitations of Generative Model**

① Training Complexity

↳ Requires huge datasets, GPU, long training time

② Quality Control

↳ Output may contain → distortions, unrealistic artifacts, inconsistencies

③ Overfitting

↳ Model memorizes training data → less diverse output, repeated samples

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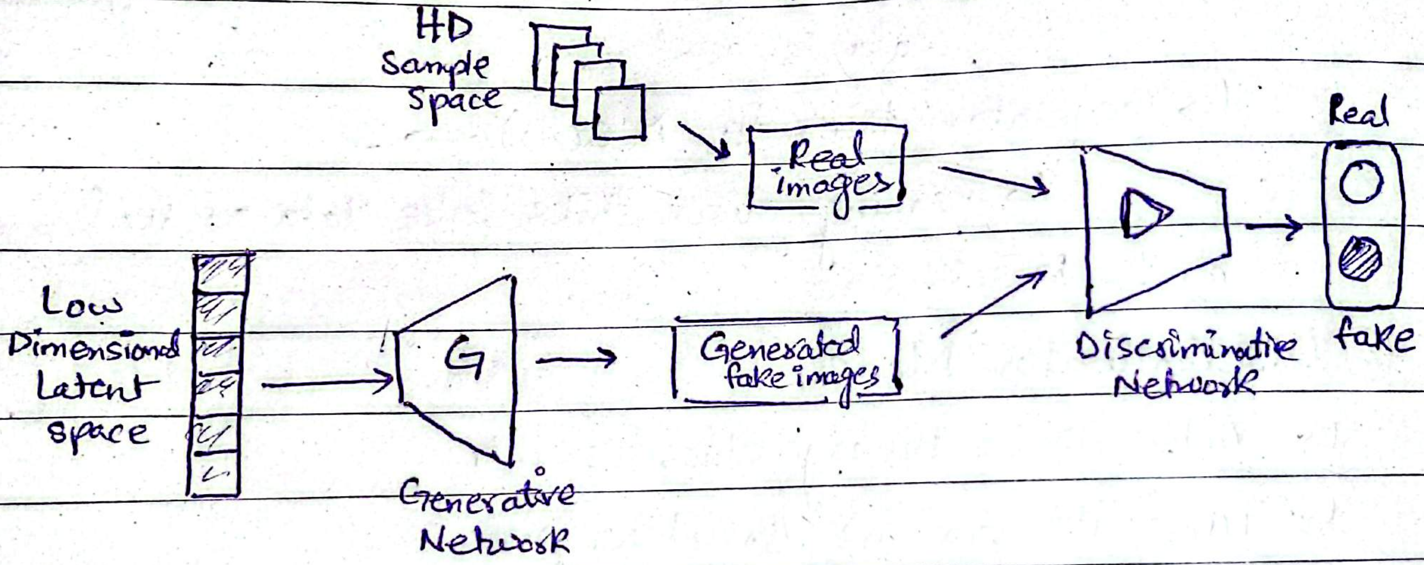
Date
Day

④ Ethical Concerns
↳ Can generate deep fakes, fake voices
⑤ Data Dependency
↳ Poor training data → poor generated output

- Generative Adversarial Network
↳ Two neural networks compete against each other



① Generator Model
↳ Deep Neural network that takes random noise as input to generate data
↳ Learns underlying data patterns, adjust internal params by backpropagation
↳ Goal: Produces samples that discriminator can't classify as real

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↳ Tries to minimize loss :

$$J_G = \frac{1}{m} \sum_{i=1}^{m} \log D(G(z_i))$$

where,

J_G = How well Generator fools discr.

$G(z_i)$ = generated sample from random noise "z_i"

$D(G(z_i))$ = Discriminator's estimated probability that data is real

↳ Generator tries to $D(G(z_i)) \longrightarrow 1$

↳ meaning discr. thiks fake data as real

m = batch size

② Discriminator Model

↳ Acts as a binary classifier,

↳ Tune its params thru backpro.

↳ For image data uses CNN to help extract features

↳ Tries to maximize loss

$$J_D = -\frac{1}{m} \sum_{i=1}^{m} \log D(x_i) - \frac{1}{m} \sum_{i=1}^{m} \log(1 - D(G(z_i)))$$

where,

J_D = how well the disc. classifies real vs fake

x_i = Real example

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Day

↳ $G(z_i)$ = fake sample from generator

$D(x_i)$ = prob that x_i is real

$D(G(z_i))$ = Disc. prob that fake sample is real

overs that a particular fake sucessfully, producing same sucessfull ... multiple

↳ Discriminator goal is to maximize

$D(x) \rightarrow$ real is real

$1 - \underbrace{D(G(z_i))}_{\text{fake is real}} \rightarrow$ fake is fake

$$\min_G \max_D (G, D) = \left[\mathbb{E}_{x \sim p_{data}}[\log D(x)] + \mathbb{E}_{z \sim p_z(z)}[\log(1 - D(G(z)))]\right]$$

↳ generater tries to minimize loss by generating samples that fool discriminator

↳ Discr. tries to maximized this loss by correct classification

↳ This is called $\boxed{\text{Adverserial training}}$

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Date

Day

↳ Tries to minimize los

$$J_G = \frac{1}{m} \sum_{i=1}^{m} \log$$

where,

J_G = How well (

$G(z_i)$ = generc

$D(G(z_i))$ = D

ssi

→ Equilibrium State

↳ Discriminator cannot distinguish b/w fake data/ real data

$$D(G(z_i)) \approx 0.5$$

→ Mode Collapse

↳ A problem in GANs where the generator starts producing only a limited variety of outputs instead of generating diverse samples

↳ Generator only learns few modes, ignores the rest, so diversity collapse

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Date ______ Day ______

- Happens Why?
  - The generator discovers that a particular fake sample fools discr. sucessfully,
  - therefore it keeps producing same sucessfull sample repeatedly.

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